Beyond Prepare-and-Measure

The Cost of Trustless Detectors in Quantum Key Distribution

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2 Sep 2026

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Motivation

Classical public-key crypto is not quantum-safe.

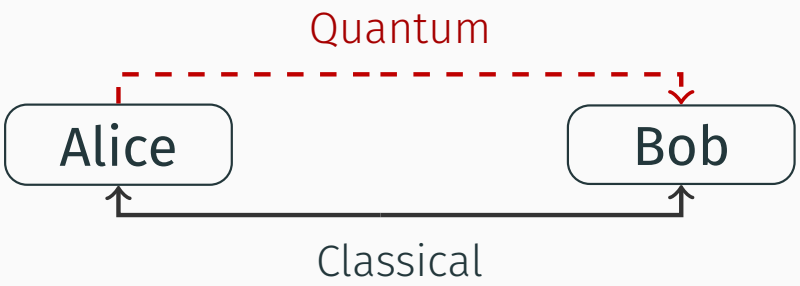
- RSA / ECC security rests on factoring and discrete log
- Shor's algorithm breaks both in polynomial time
- QKD sidesteps this: security from physics, not complexity

QKD has matured – national fibre networks, metropolitan testbeds.

Providers – which protocol to deploy, and how to configure?

- BB84 (1984): the original Prepare-and-Measure scheme
- MDI-QKD (2012): removes the need to trust the detector

Two Protocols



BB84 – trusted detector



MDI-QKD – untrusted relay

Charlie performs BSM on incoming photons and announces *correlation* – adversary learns nothing. Keys require *both* photons to survive.

Both protocols bound by the same physics: key rate decays exponentially with distance (PLOB bound, Pirandola et al. 2017).

Methodology Overview

Three comparison axes, one simulation engine, one cost model.

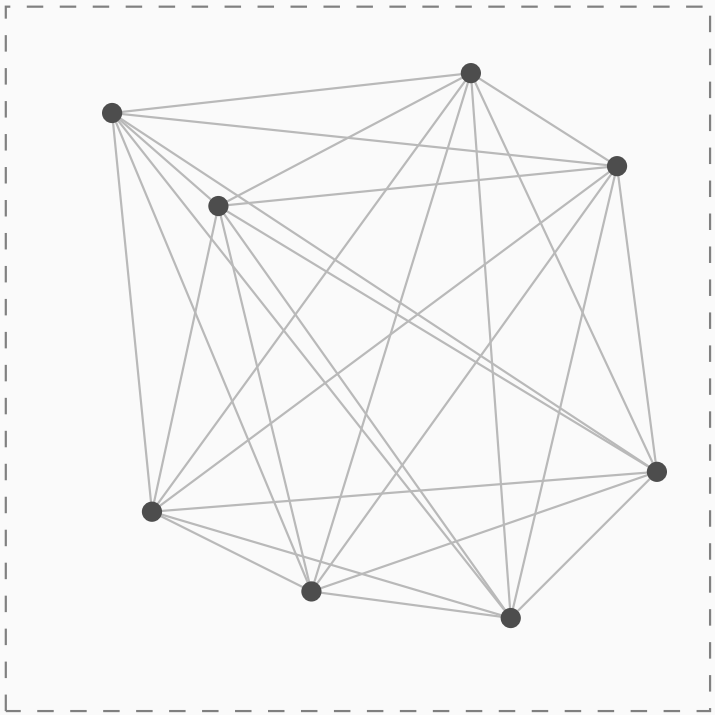
Performance
Sifted key rate vs.
10-layer hardware model

Cost
Symbolic CapEx,
per-component prices

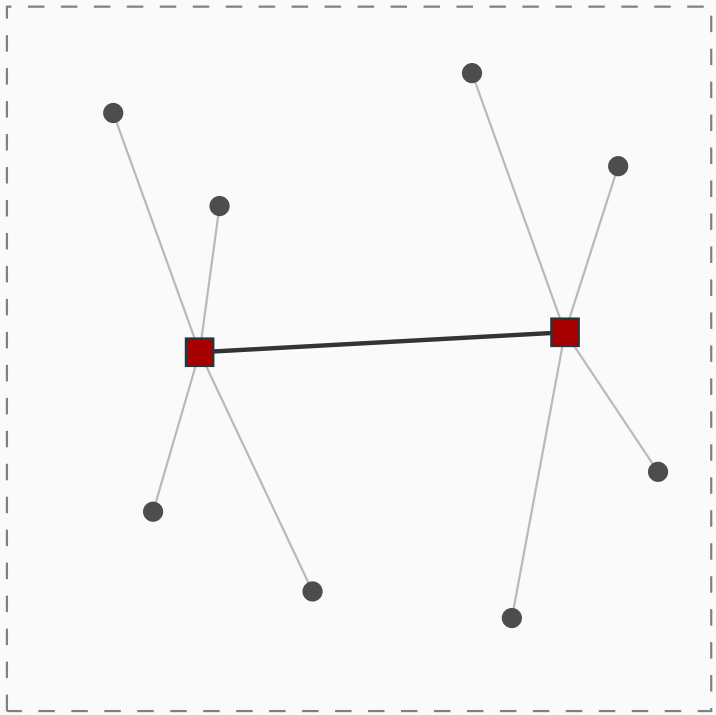
Scalability
Marginal rate as topology
parameters vary

Each hardware configuration is run as Monte Carlo trials with NetSquid to obtain mean key rates and their spread.

Network Topologies



BB84 direct mesh: $\binom{N}{2} = \mathcal{O}(N^2)$ links



MDI star-mesh: $\mathcal{O}(N)$ spokes + $\mathcal{O}(K^2)$ backbone

Same N=8-user layout under each protocol (K=2 relays, right). Relay position minimises total fibre length.

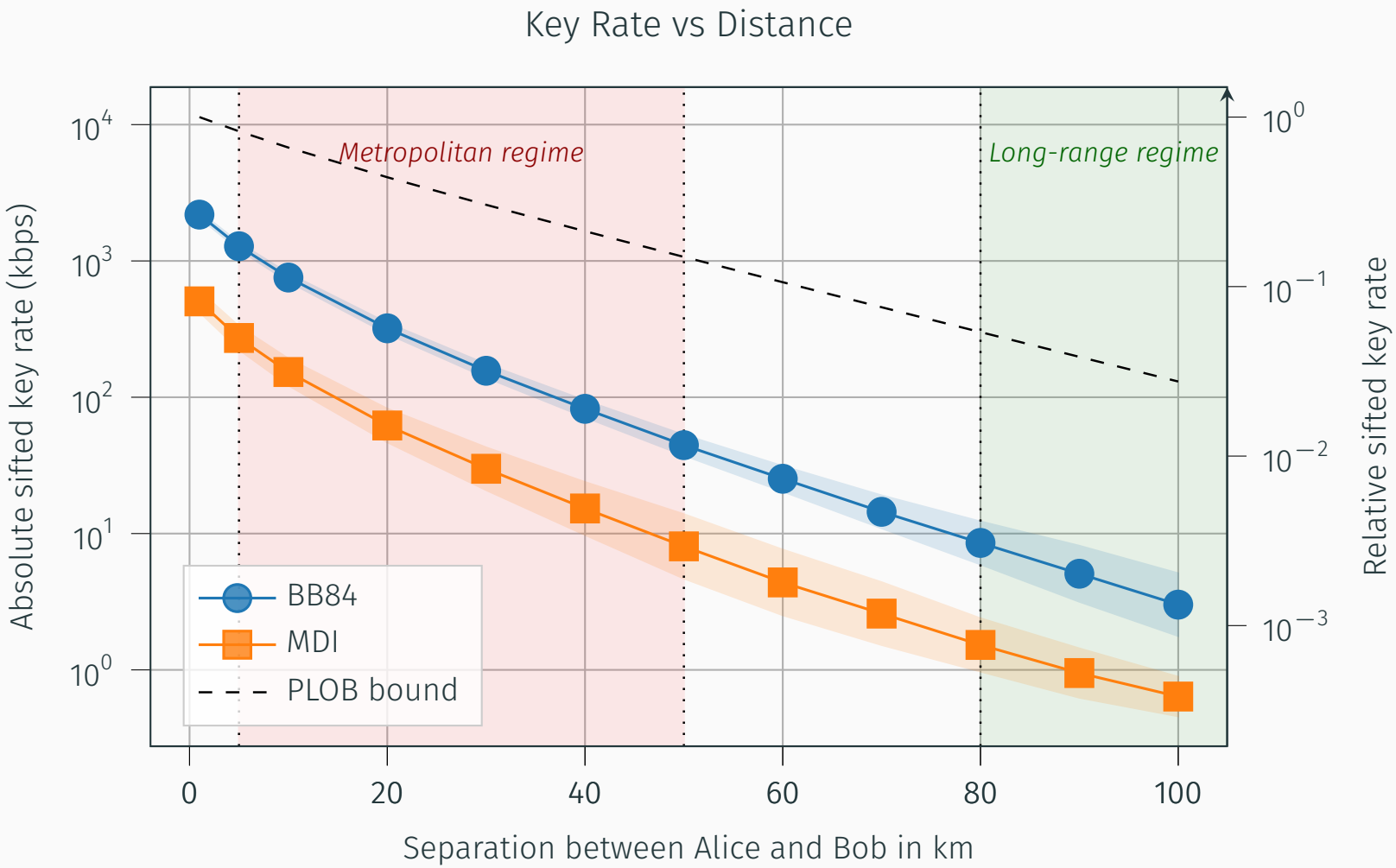
The Cost Model

Symbolic, per-component CapEx: c_s (source), c_d (detector), c_f (fibre/km).

Component	BB84	MDI
Single-photon sources	N	N
Single-photon detectors	$2N$	$4K$
Fibre links	$N(N-1)/2$	$N + K(K-1)/2$

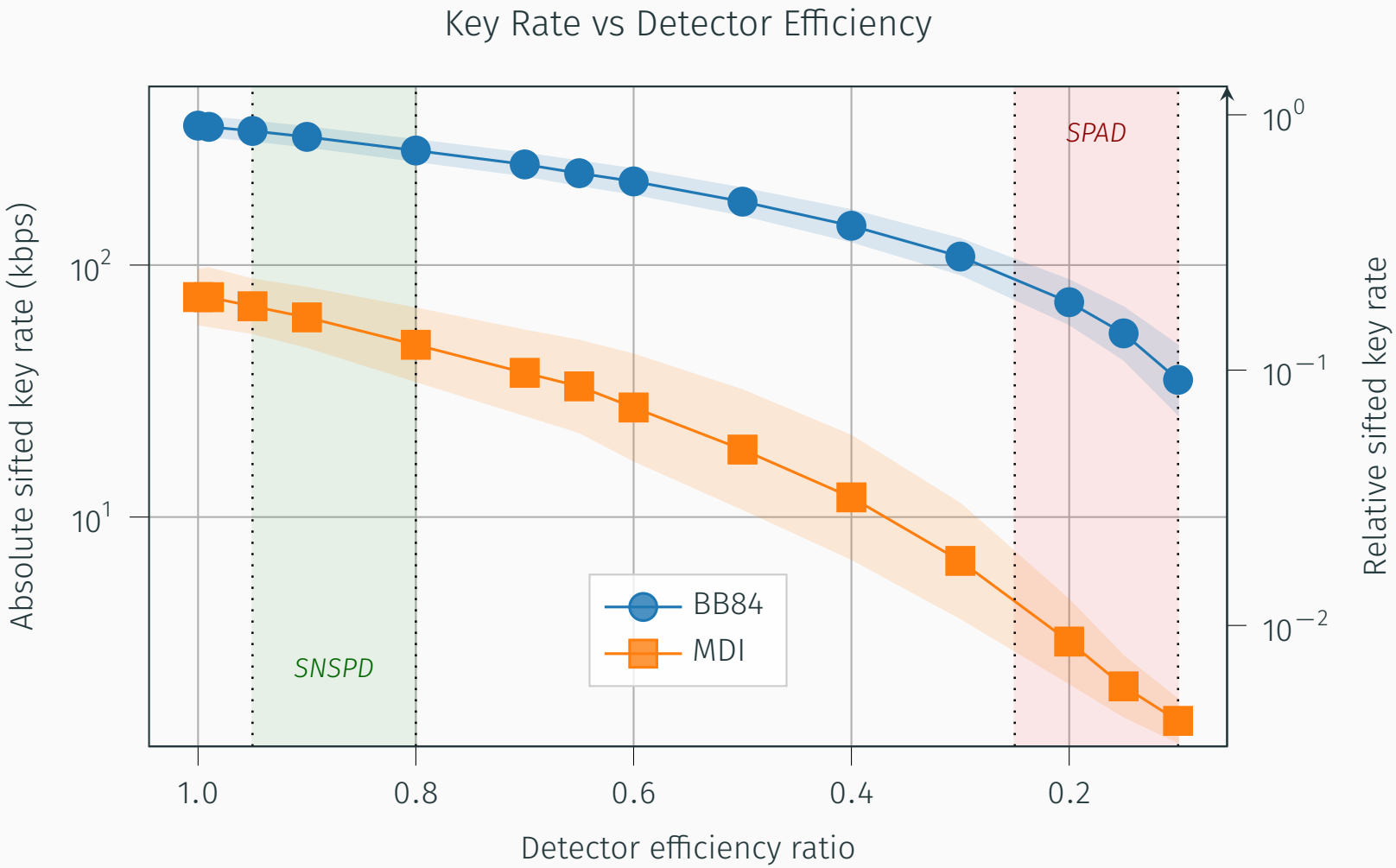
BB84's fibre term is $\mathcal{O}(N^2)$; MDI's is $\mathcal{O}(N)$ at fixed $K \ll N$.

P2P Result: Key Rate vs. Distance



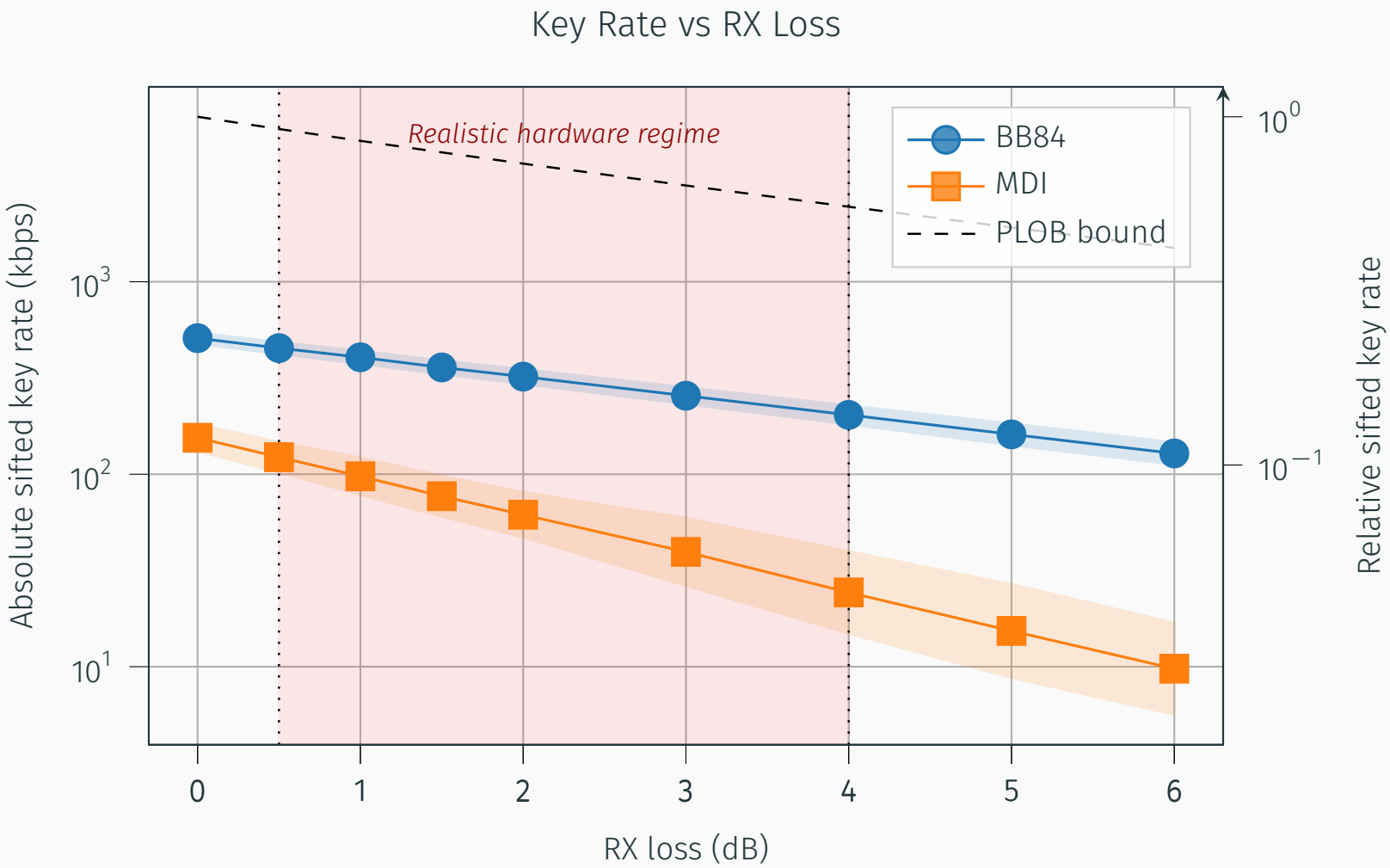
BB84/MDI ratio around 5× across 1–100 km, even as rate falls.

P2P Result: Detector Efficiency Widens the Gap



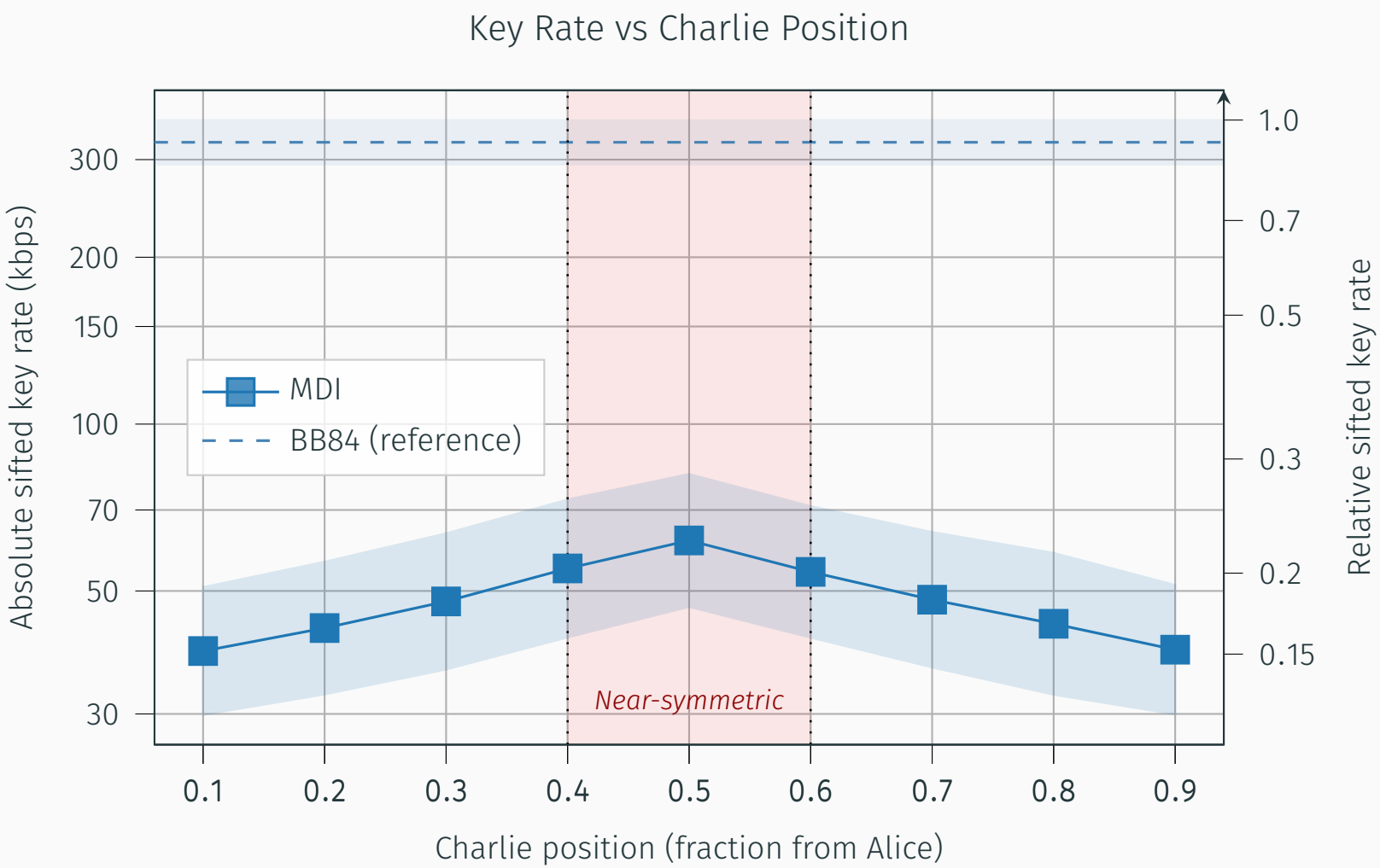
MDI's success probability scales as η^2, BB84's as η. Ratio widens: $\sim 5\times$ to $25\times$.

P2P Result: Receiver Loss Widens the Gap



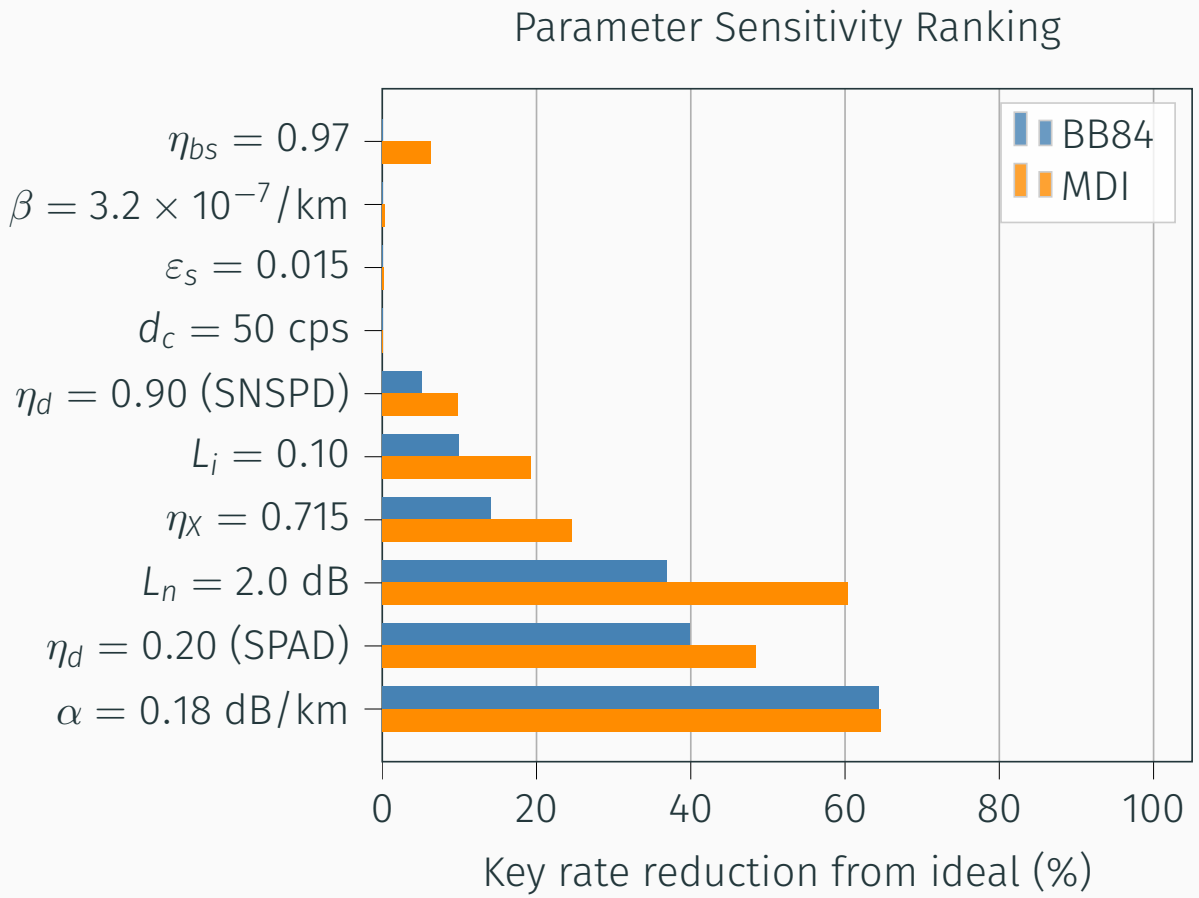
Per-arm mechanism, as with efficiency. Ratio widens from ∼5× to 13×.

P2P Result: Relay Placement Matters



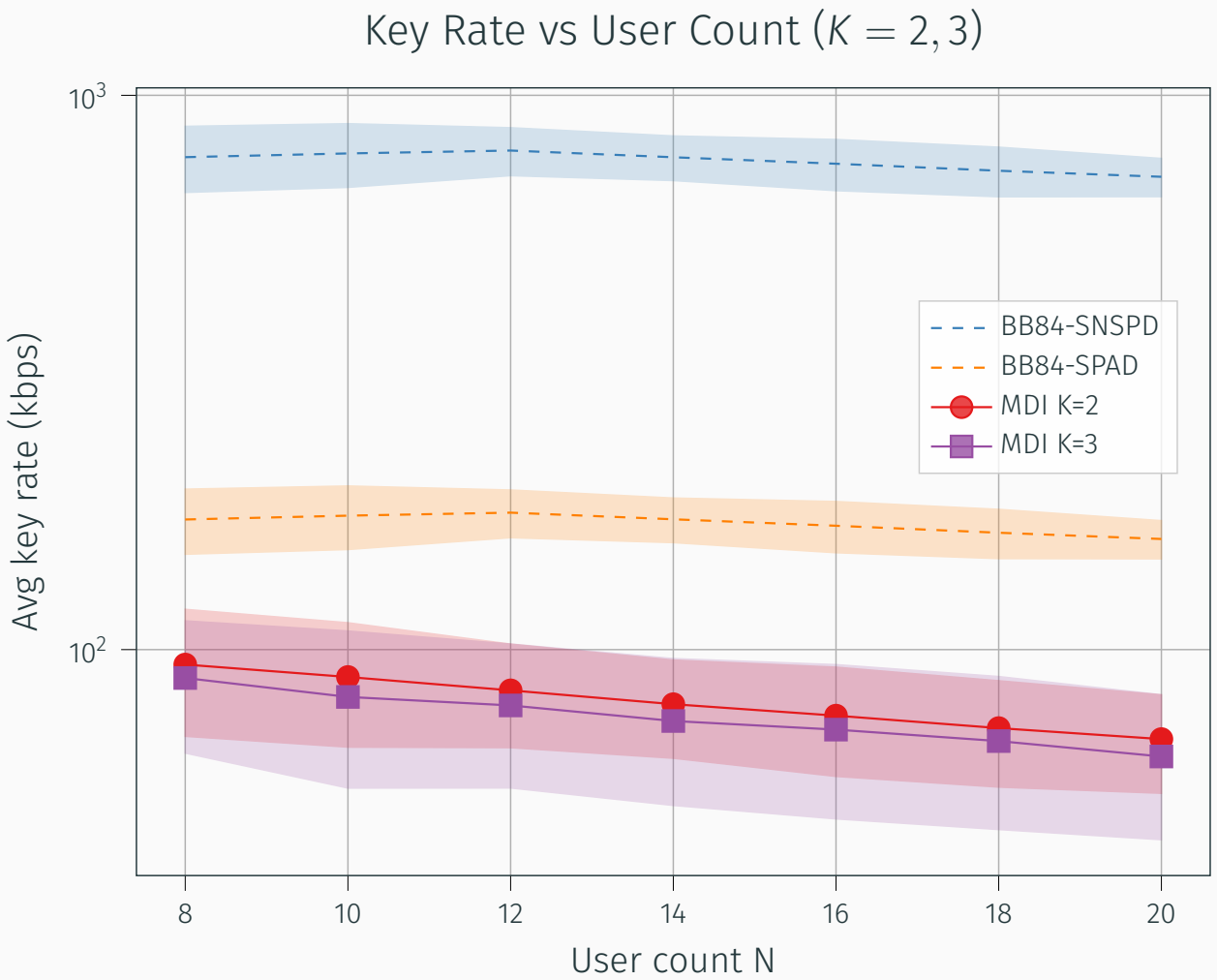
MDI peaks at the symmetric midpoint (61.7 kbps). Off-centre widens the BB84/MDI ratio to 8.2–8.3×.

P2P Result: What Matters Most?



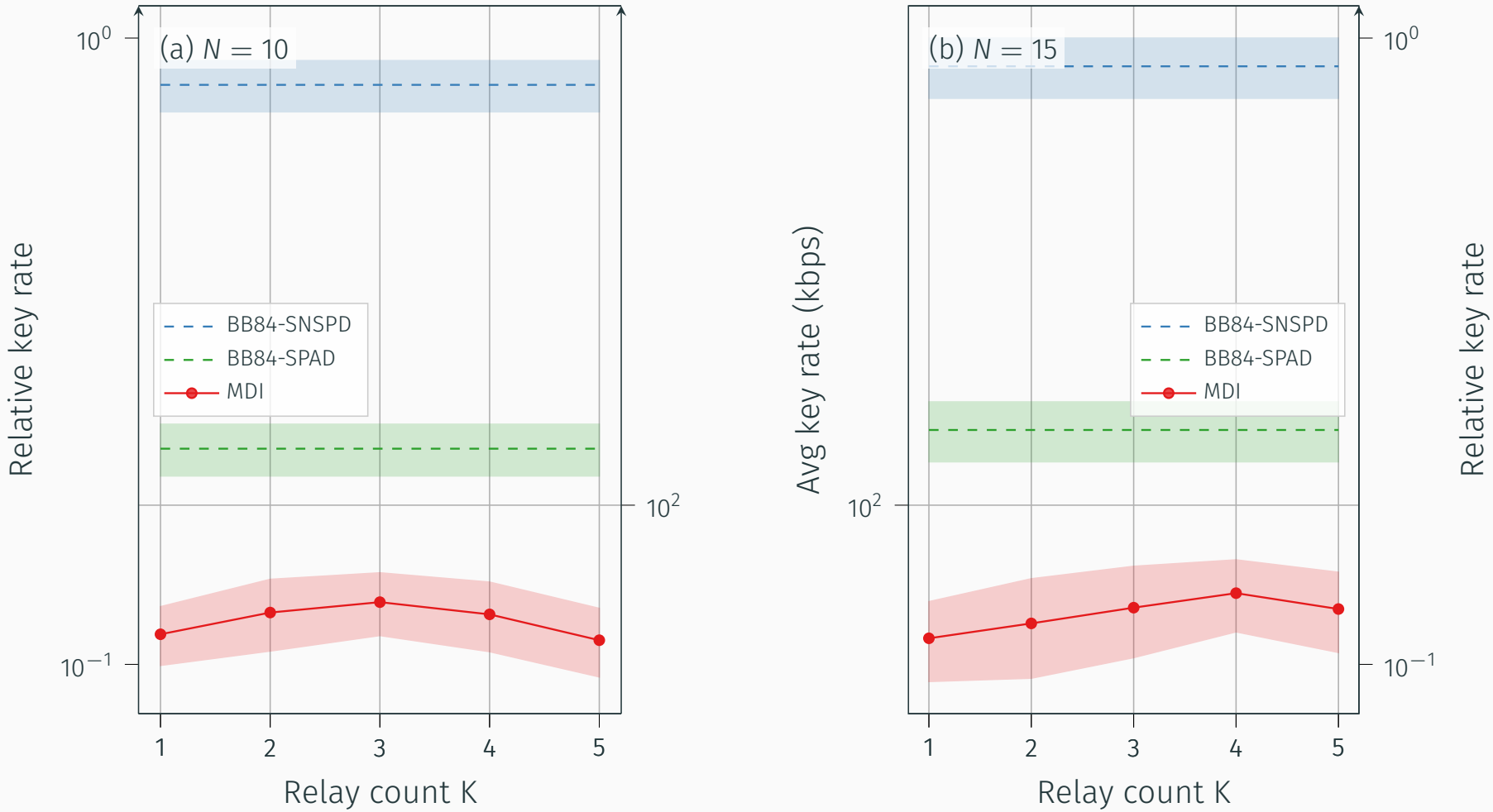
Fibre attenuation dominates both protocols almost identically (∼64%).
Beyond that, every per-arm parameter hurts MDI 1.6–2× more than BB84.

Result: Growing Into a Network



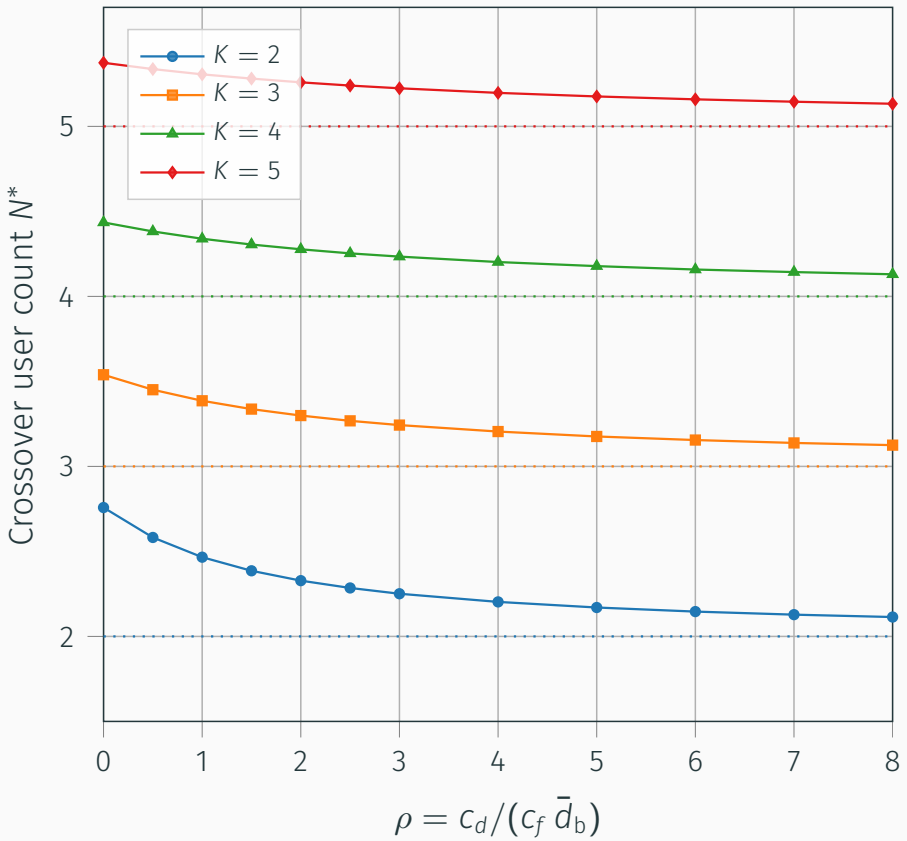
Fixed relay infrastructure, organic user growth: MDI's rate falls ∼27% as N:8→20; BB84 changes under 8%.

Result: How Many Relays?



Key rate peaks at K=3 (N=10) and K=4 (N=15) – non-monotonic, $\approx \mathcal{O}(\sqrt{N})$.
The rate-optimal K is not the fibre-cost-optimal K.

Result: The Cost Crossover



Assume symmetric detector pricing across protocols.

$$N^*(\rho, K) \in (K,\ K{+}1)$$

for *any* price ratio $\rho = c_d/c_F$.

MDI is cheaper at every realistic (N, K), cost per new user is *constant*; BB84's grows linearly.

What Are You Trusting?

A customer's threat model fixes a trust level.

Level	Trusts	Protocol
0	Nothing	DI-QKD
1	Source only	**MDI-QKD**
2	Source + detector	**BB84**
3	Source + detector + relay	TBB84

A Level-1 customer has **no BB84 substitute**, at any price or rate.

Future Work

To deepen our understanding of protocol asymmetries we should improve:

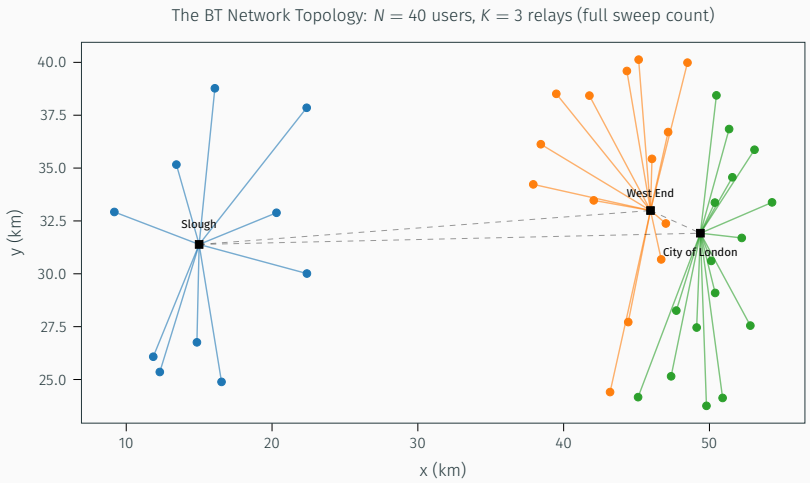
- **Physics** – finite-key corrections, decoy-state sources, secure (not just sifted) key rate
- **Cost model** – real vendor prices, operational expenditure, asymmetric detector pricing
- **Scale** – quantum memories + repeaters, live traffic contention, further real-world topologies
- **Tool** – toward a general-purpose QKD network simulator

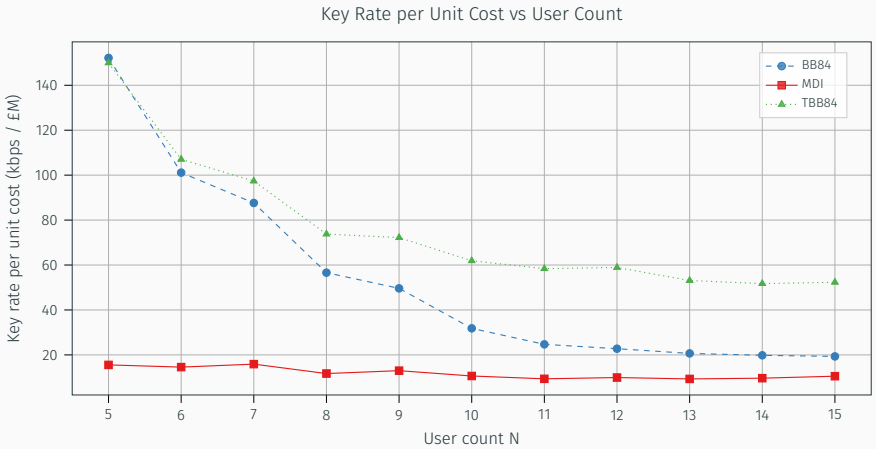
Conclusion

- Built full simulation + cost-modelling pipeline for BB84 vs. MDI-QKD
- Rate gap: $\sim 5\times$ under channel loss, $>25\times$ under per-arm hardware loss
- Cost: MDI is conditionally cheaper, regardless of detector price
- The choice is ultimately a **security** question, not a cost one
- Infrastructure built to generalise

Thank you!

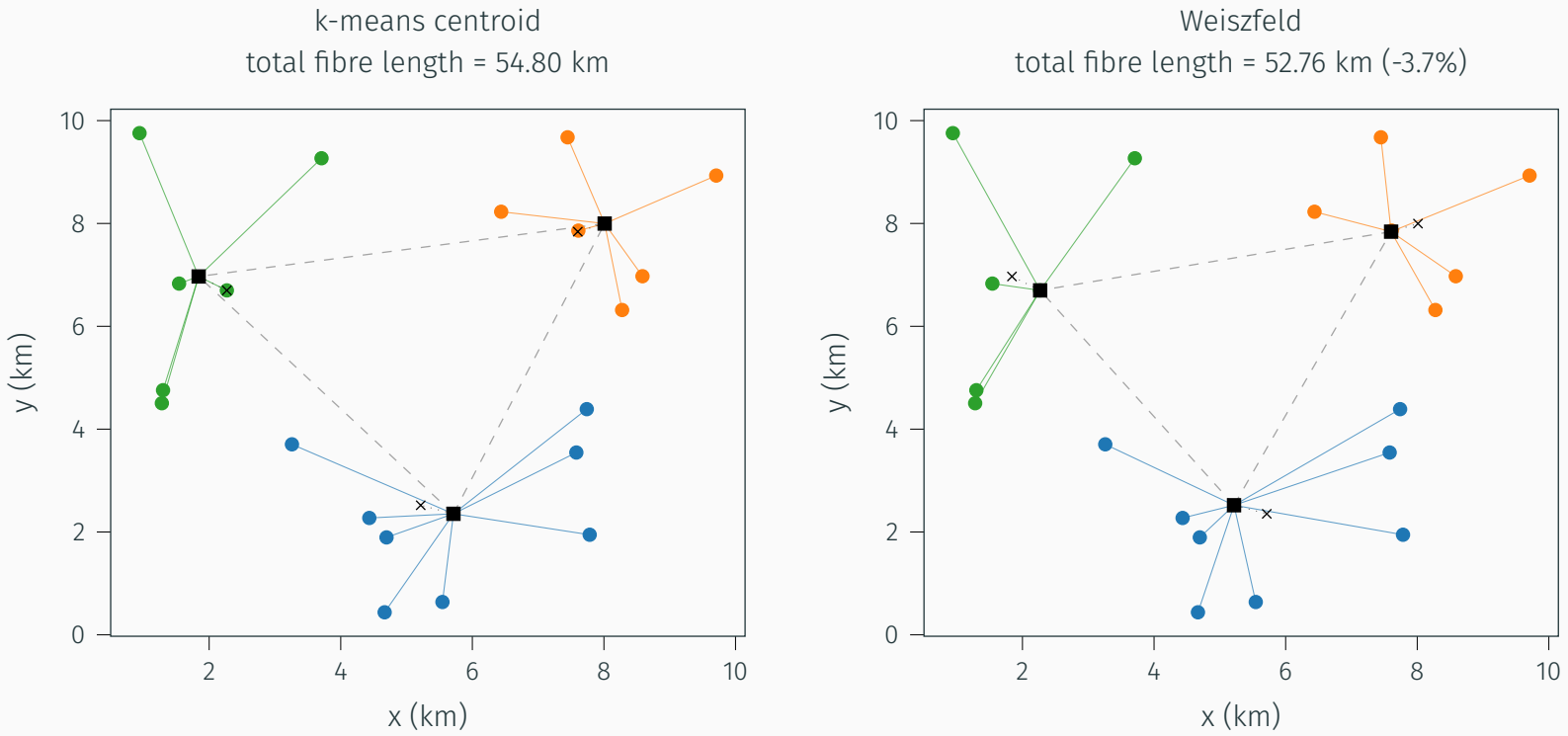
Backup: BT Network Case Study





Three real BT/Toshiba London sites (Slough, West End, City), K=3, synthetic users. Cost-efficiency ranks inverse to trust: TBB84 > BB84 > MDI, by ∼10× ($N \geq 6$) – mirrors the synthetic-topology results.

Backup: Relay Placement – Weiszfeld vs. k-means



Relay positions set by a generalised Weiszfeld fixed-point update, minimising total fibre length (spoke + backbone) – not k-means' sum-of-squares. Reduction grows with K, shrinks with N: 4.6% (K=2) → 45.6% (K=5) at N=10; only 0.6%–5.7% at N=50. Convergence guaranteed by convexity (up to 146 iterations, worst case).